VINAY.T

LINUX Interview Style Q&A with Practical Context DAY 13

Advanced Networking

100. Q: In Linux, how can you configure a network interface with a specific IP address?
A: You can assign an IP address using the ip command:
sudo ip addr add 192.168.0.10/24 dev eth0

This assigns the IP 192.168.0.10 with a subnet mask of /24 to the eth0 interface.

Note: This change is temporary and will be lost after a reboot unless configured in network files (e.g., Netplan, NetworkManager, or /etc/network/interfaces).

101. Q: How would you enable or disable a network interface in Linux?
A: To bring an interface up (enable):
sudo ip link set eth0 up

To bring it down (disable):
sudo ip link set eth0 down

This is commonly used when restarting interfaces or performing diagnostics.

102. Q: What command do you use to test DNS resolution for a domain name?
A: Use the nslookup or dig command:
nslookup example.com
or:
dig example.com

These tools help confirm whether a domain name is being resolved correctly and what nameservers are being used.

103. Q: How can you view ARP table entries in a Linux system?
A: You can display the ARP cache using:
arp -a

This shows the IP addresses and their corresponding MAC addresses that the system has communicated with.

104. Q: How would you allow traffic on a specific TCP port using iptables?
A: To allow incoming TCP traffic on port 443 (HTTPS), use:
sudo iptables -A INPUT -p tcp --dport 443 -j ACCEPT

- -A INPUT: appends the rule to the INPUT chain
- -p tcp: specifies TCP protocol
- --dport 443: designates port
- -j ACCEPT: action to allow the traffic

105. Q: After adding iptables rules, how do you ensure they persist after a reboot?
A: Use the following command to save current rules:
sudo iptables-save > /etc/iptables/rules.v4
Or, if using the iptables-persistent service:
sudo netfilter-persistent save

Persistence is crucial because iptables rules are reset on reboot unless saved.

106. Q: How do you check which ports are currently open and listening on your system?
A: Use:
ss -tuln
Explanation:

- -t: TCP
- -u: UDP
- -l: listening ports
- -n: numeric output (no DNS resolution)

This helps identify active services and their respective ports.

107. Q: How can you trace the path that packets take to reach a specific host?
A: Use the traceroute command:
traceroute example.com

This outputs each router/hop a packet crosses to reach the destination.
Useful for diagnosing network latency and routing issues.

108. Q: How would you capture network packets for analysis?
A: Use tcpdump, a powerful packet analyzer:
sudo tcpdump -i eth0

To filter for HTTP traffic:
sudo tcpdump -i eth0 port 80

This helps monitor traffic on specific interfaces and ports.

109. Q: How do you manually map hostnames to IP addresses in Linux?
A: Edit the /etc/hosts file using a text editor:
sudo nano /etc/hosts

Add a line like:
192.168.0.10   dev.example.local

This maps the domain dev.example.local to the IP 192.168.0.10. The system will check this file before querying DNS.

Scripting and Automation

110. Q: How do you create a simple shell script in Linux?
A: You start by creating a file with a .sh extension and include the shebang at the top to define the interpreter:

#!/bin/bashecho "Hello, World!"

- The line #!/bin/bash tells the system to use the Bash shell to run the script.
- Save this script with any name, e.g., hello.sh.

111. Q: How can you make a shell script executable?
A: Use the chmod command:
chmod +x script_name.sh

This grants execute permission to the script so you can run it directly using ./script_name.sh.

112. Q: How do you run a shell script in Linux?
A:
There are multiple ways:

1. If executable:
```
./script_name.sh
```

2. Using the shell explicitly:
```
bash script_name.sh
```

113. Q: How are variables used in shell scripts?
A: Variables are defined without spaces:
```
name="DevOps"
```

To access the value:
```
echo "Welcome to $name"
```

Note: Use double quotes when referencing variables to avoid word splitting or globbing issues.

114. Q: How can you take input from the user in a script?
A: Use the read command:
```
echo "Enter your name:"
read username
echo "Hello, $username!"
```

This pauses script execution and waits for user input.

115. Q: How do you pass and access arguments in a shell script?
A: Shell scripts accept positional parameters:

- $0 – script name
- $1, $2, ... – arguments passed

Example:
```
#!/bin/bash
echo "First argument: $1"
echo "Second argument: $2"
```

Run it as:
```
./script.sh arg1 arg2
```

116. Q: How do you write an if-else condition in a shell script?
A:
```
#!/bin/bash
if [ "$1" -gt 10 ]; then
  echo "Number is greater than 10"
else
  echo "Number is 10 or less"
fi
```

This checks if the first argument is greater than 10. Always quote variables to avoid unexpected behavior with empty strings.

117. Q: How do you write a for loop in a Bash script?
A: Here's a simple loop from 1 to 5:

```
for i in {1..5}; do
  echo "Iteration $i"
done
```

This loop is useful for iterating over ranges, files, or command output.

118. Q: How can you schedule a shell script using cron?
A:Edit the crontab:
crontab -e

Add a cron expression:
0 6 * * * /path/to/script.sh

This example runs the script daily at 6:00 AM.

Make sure your script has executable permission and uses full paths to commands and files.

119. Q: How do you debug a shell script during execution?
A: Use the -x option with bash:
bash -x script.sh

This prints each command and its arguments as they are executed—very helpful for troubleshooting complex scripts.

Advanced Topics

120. Q: What is SELinux and why is it important?
A:SELinux (Security-Enhanced Linux) is a security module in Linux that provides mandatory access control (MAC). It defines access policies beyond traditional file permissions, enhancing system security by restricting how processes and users can access files, sockets, and other resources.

- Useful in high-security environments like enterprise and government systems.

121. Q: How can you check the current status of SELinux on a system?
A: You can run the following command:
sestatus

It will return the current SELinux mode: enforcing, permissive, or disabled.

122. Q: How do you change the SELinux mode on a running system?
A: Use the setenforce command:
setenforce 0 # Changes to permissive mode
setenforce 1 # Changes to enforcing mode

Note: This change is temporary. To make it permanent, edit /etc/selinux/config.

123. Q: How do you list all running containers in Docker?
A: Use:
docker ps

This displays all currently running containers along with their container ID, image, status, and ports.

124. Q: How do you stop a running Docker container?
A: Run:
docker stop <container_id>

You can get the container ID from docker ps. This sends a SIGTERM to the container and gives it time to shut down gracefully.

125. Q: How do you generate an SSH key pair for secure login?
A: Use:
ssh-keygen

- This creates a public-private key pair in ~/.ssh/.
- Public key: id_rsa.pub
- Private key: id_rsa
- The public key should be copied to the remote server's ~/.ssh/authorized_keys file.

126. Q: How do you use rsync for efficient file transfers?
A: The basic syntax:
rsync -avz source/ user@remote:/destination/

Flags:
- -a: archive mode (preserves permissions, timestamps, etc.)
- -v: verbose
- -z: compress data during transfer

Rsync only transfers changed files, making it highly efficient.

127. Q: How can you check the default gateway of your system?
A: Run:
ip route

The line that starts with default via shows the default gateway.

128. Q: How do you monitor file changes in real-time in Linux?
A: Use the inotify-tools package. Example command:
inotifywait -m /path/to/watch

- The -m option puts it in monitoring mode.
- Useful for real-time file auditing or triggering automation scripts.

129. Q: How do you create and activate a Python virtual environment?
A: To create:
python3 -m venv myenv

To activate:
source myenv/bin/activate

This isolates Python dependencies per project and prevents version conflicts.

Disk Management

130. Q: How can you check the current swap usage on a Linux system?
A: You can use one of the following commands:
swapon --show
or
free -h

- swapon --show displays details about active swap partitions and files.
- free -h shows memory and swap usage in a human-readable format.

131. Q: How do you list all disk partitions available on a Linux system?
A: Use: lsblk

This command shows block devices and their mount points, useful for visualizing partition layouts.

132. Q: How do you format a disk or partition to the ext4 filesystem?
A: Run:
mkfs.ext4 /dev/sdX

Replace /dev/sdX with the actual disk or partition name.

⚠ Warning: This will erase all data on the specified disk/partition.

133. Q: How do you create a swap partition?
A: Use the following steps:
mkswap /dev/sdX   # Format the partition as swap
swapon /dev/sdX   # Enable the swap space

Optionally, add an entry to /etc/fstab for persistence after reboot.

134. Q: How do you check disk usage across mounted filesystems?
A: Use: df -h

- df stands for disk free.
- The -h flag makes output human-readable (GB, MB, etc.).

135. Q: How can you check inode usage on the filesystem?
A: Run: df -i

This shows inode counts and usage per mount point, which is useful when disk space seems free but the system reports "no space left" due to inode exhaustion.

136. Q: How do you mount a disk or partition to a directory?
A: Use:
mount /dev/sdX /mount_point

Make sure the mount point (directory) exists before running this command.

137. Q: How do you unmount a disk from a mount point?
A: Run:
umount /mount_point

Ensure no processes are using the mount point, or use lsof to identify open files.

138. Q: How can you resize a disk partition?
A: Steps generally include:
1. Modify the partition using tools like fdisk or parted.
2. Resize the filesystem using:
   resize2fs /dev/sdX

- Only supported for certain filesystems like ext4.
- Always back up data before resizing.

139. Q: How do you check the SMART status of a hard drive?
A: Use:
smartctl -a /dev/sdX
- Part of the smartmontools package.
- Displays detailed SMART health and diagnostic information for drives.

Process Management

140. Q: How do you list all running processes on a Linux system?
A: You can use:
ps aux
- a – Lists processes from all users.
- u – Displays the user/owner of each process.
- x – Includes processes not attached to a terminal (e.g., daemons).

This command gives a comprehensive overview of all running processes.

141. Q: How do you terminate a process using its PID?
A: Use the kill command followed by the Process ID (PID):
kill PID

This sends the default SIGTERM signal (15), which asks the process to terminate gracefully.

142. Q: How do you kill a process by its name instead of PID?
A: Use the pkill command:
pkill process_name

This will send a signal (default is SIGTERM) to all processes matching the name.

143. Q: How can you stop (pause) a running process?
A: You can use:
kill -STOP PID

This sends the SIGSTOP signal to suspend the process. It's similar to pressing Ctrl+Z in the terminal.

144. Q: How do you resume a stopped process?
A: Use:
kill -CONT PID

This sends the SIGCONT signal to continue a process previously stopped with SIGSTOP.

145. Q: How do you view active processes in real-time?
A: Use either:
top
or, if available:
htop

- top shows CPU/memory usage, PID, user, command, and more in real time.
- htop is an interactive and user-friendly alternative with color-coding and filtering options.

146. Q: How can you check which processes are consuming the most memory?
A: Sort the output of ps by memory usage:
ps aux --sort=-%mem

This lists processes from highest to lowest memory consumption.

147. Q: How can you check which processes are consuming the most CPU?
A: Sort the output of ps by CPU usage:
ps aux --sort=-%cpu

Useful for identifying CPU-hogging applications.

148. Q: How do you check the status of a specific process?
A: Use:
ps -o stat= -p PID

This displays the process status, which can be:
- R (running),
- S (sleeping),
- T (stopped),
- Z (zombie), etc.

149. Q: How do you find the PID(s) of a process by its name?
A: Use:
pgrep process_name

This returns the PID(s) of all processes matching the specified name.

150. Q: How can you count the number of currently running processes?
A: You can use:
ps aux | wc -l

This counts the number of lines (processes) listed by ps.